

Lab 7: PMOS

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab is to measure the I_D - V_{SD} characteristics of a PMOS transistor and to develop a simple PMOS model using the threshold voltage V_{tp} , the transconductance parameter k_p , and the channel-length modulation parameter λ [1]. The PMOS device was taken from a CD4007UB CMOS transistor array, using pin 6 as the gate, pin 13 as the drain, and pin 14 as the source [3]. During the experiment, V_{SG} was fixed at 3 V, and V_{SD} was varied from 0.5 V to 5.0 V.

2) Equipment Used

Table 1: Equipment used in Lab 7

Item	Quantity
CD4007UB CMOS transistor array	1
Resistor, approximately 1 k Ω	1
DC Power Supply	1
Digital Multimeter	1
Breadboard and wires	1 set

3) Theory

3.1) PMOS Operation Regions

For an enhancement-type PMOS transistor, the transistor is turned on when the source-gate voltage is larger than the magnitude of the threshold voltage:

$$V_{SG} > |V_{tp}|$$

The overdrive voltage is defined as

$$|V_{OV}| = V_{SG} - |V_{tp}|$$

The operation region is determined by comparing V_{SD} with $|V_{OV}|$. If

$$V_{SD} < |V_{OV}|$$

the transistor operates in the triode region. The drain current is modeled as

$$I_D = k_p \left[(V_{SG} - |V_{tp}|)V_{SD} - \frac{1}{2}V_{SD}^2 \right]$$

If

$$V_{SD} \geq |V_{OV}|$$

the transistor operates in the saturation region. With channel-length modulation included, the drain current is modeled as

$$I_D = \frac{1}{2}k_p(V_{SG} - |V_{tp}|)^2(1 + \lambda V_{SD})$$

where λ represents the channel-length modulation effect. In a real MOSFET, the saturation current is not perfectly constant. As V_{SD} increases, the effective channel length changes slightly, so I_D continues to increase.

3.2) Current Measurement Method

The drain current was obtained from the voltage drop across the drain resistor. According to the lab instruction, the current from Action #1 is calculated by

$$I_{D1} = \frac{|V_{SS}| - |V_{SD}|}{R_D}$$

where R_D is the measured drain resistor value. In this experiment,

$$R_D = 989.5 \, \Omega$$

The calculated current from the PMOS model is denoted as I_{D2} .

The percentage error is calculated by

$$\text{Error} = \left| \frac{I_{D2} - I_{D1}}{I_{D1}} \right| \times 100\%$$

4) Experimental Setup

The PMOS transistor was connected using the CD4007UB chip. The source terminal was connected to the reference node. The gate terminal was connected to a fixed negative DC voltage so that $V_{SG} = 3\text{ V}$. The drain terminal was connected to the drain resistor R_D , and the other side of R_D was connected to the adjustable negative DC supply V_{SS} . By changing V_{SS} , different values of V_{SD} were obtained.

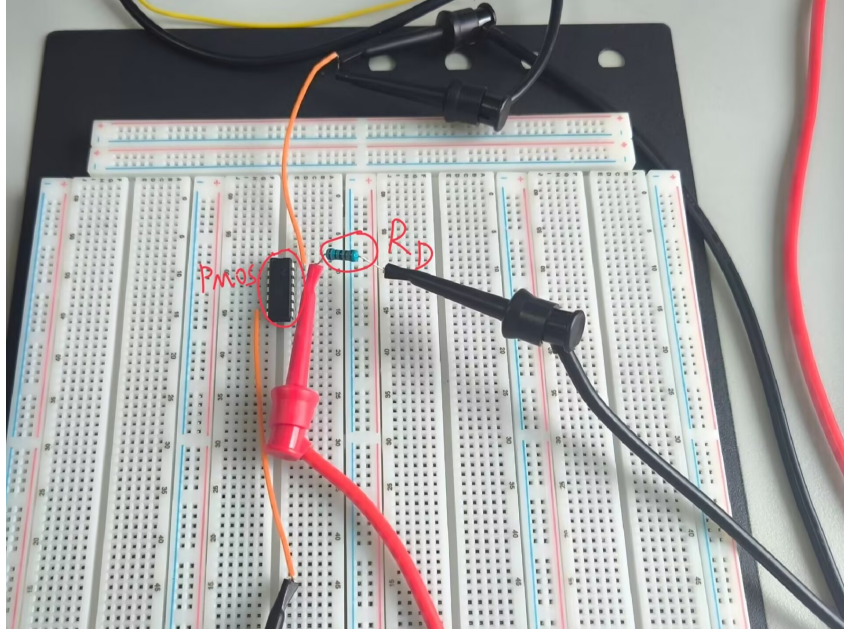


Figure 1: Experimental setup for measuring the PMOS I-V characteristics

5) Experimental Data

The measured value of the drain resistor was

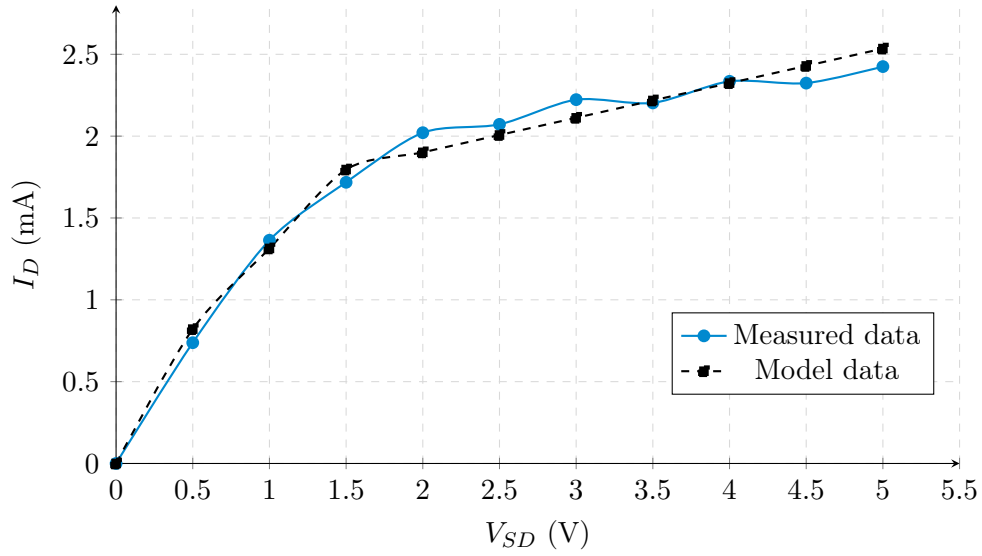
$$R_D = 989.5 \, \Omega$$

The measured and calculated current values are shown in Table 2. The measured current I_{D1} was calculated from the recorded V_{SS} , V_{SD} , and R_D values.

Table 2: Measured and calculated PMOS drain current

V_{SS} [V]	V_{SD} [V]	I_D from Action #1	I_D from Action #2	Error [%]
-1.23	0.5	7.38×10^{-4}	8.21×10^{-4}	11.35
-2.35	1.0	1.364×10^{-3}	1.314×10^{-3}	3.66
-3.20	1.5	1.718×10^{-3}	1.796×10^{-3}	4.51
-4.00	2.0	2.021×10^{-3}	1.901×10^{-3}	5.94
-4.55	2.5	2.072×10^{-3}	2.007×10^{-3}	3.14
-5.20	3.0	2.223×10^{-3}	2.112×10^{-3}	4.99
-5.68	3.5	2.203×10^{-3}	2.218×10^{-3}	0.68
-6.31	4.0	2.335×10^{-3}	2.324×10^{-3}	0.46
-6.80	4.5	2.324×10^{-3}	2.429×10^{-3}	4.51
-7.40	5.0	2.425×10^{-3}	2.535×10^{-3}	4.51
Average Error				4.38

The measured and calculated I_D - V_{SD} curves are shown in Fig. 2.

**Figure 2:** Measured and calculated I_D - V_{SD} characteristics of the PMOS for $V_{SG} = 3$ V

6) Model Extraction

6.1) Determination of $|V_{tp}|$, k_p , and λ

The PMOS model was developed using the standard long-channel MOSFET equations [2]. In this lab,

$$V_{SG} = 3.00 \text{ V}$$

The threshold voltage magnitude was selected as

$$|V_{tp}| = 1.50 \text{ V}$$

Therefore, the overdrive voltage is

$$|V_{OV}| = V_{SG} - |V_{tp}| = 3.00 - 1.50 = 1.50 \text{ V}$$

Thus, the boundary between the triode region and the saturation region is around

$$V_{SD} = 1.50 \text{ V}$$

For $V_{SD} < 1.50 \text{ V}$, the triode-region equation was used. For $V_{SD} \geq 1.50 \text{ V}$, the saturation-region equation with channel-length modulation was used.

The saturation-region equation is

$$I_D = \frac{1}{2}k_p|V_{OV}|^2(1 + \lambda V_{SD})$$

Two saturation-region data points were used to estimate λ :

$$V_{SD1} = 1.50 \text{ V}, \quad I_{D1} = 1.718 \times 10^{-3} \text{ A}$$

$$V_{SD2} = 5.00 \text{ V}, \quad I_{D2} = 2.425 \times 10^{-3} \text{ A}$$

Taking the ratio of these two currents gives

$$\frac{I_{D1}}{I_{D2}} = \frac{1 + \lambda V_{SD1}}{1 + \lambda V_{SD2}}$$

Substituting the measured data:

$$\frac{1.718 \times 10^{-3}}{2.425 \times 10^{-3}} = \frac{1 + 1.5\lambda}{1 + 5\lambda}$$

Solving this equation gives

$$\lambda = 0.142857 \text{ V}^{-1}$$

Then, using the average saturation current and average saturation voltage,

$$I_{D,\text{avg}} = 2.165 \times 10^{-3} \text{ A}$$

$$V_{SD,\text{avg}} = 3.25 \text{ V}$$

the value of k_p was obtained from

$$k_p = \frac{2I_{D,\text{avg}}}{|V_{OV}|^2(1 + \lambda V_{SD,\text{avg}})}$$

Substituting the values:

$$k_p = \frac{2(2.165 \times 10^{-3})}{(1.5)^2(1 + 0.142857 \times 3.25)}$$

Therefore,

$$k_p = 1.314 \times 10^{-3} \text{ A/V}^2$$

6.2) Extracted Model Parameters

Table 3: Extracted PMOS model parameters

Parameter	Value
R_D (Ω)	989.5
$ V_{tp} $ (V)	1.50
$ V_{OV} $ (V)	1.50
k_p (A/V ²)	1.314×10^{-3}
λ (V ⁻¹)	0.142857

7) Discussion

The measured I_D – V_{SD} curve shows the expected PMOS behavior. When V_{SD} is small, the current increases quickly with V_{SD} , which corresponds to the triode region. When V_{SD} becomes larger than about 1.5 V, the transistor enters the saturation region.

However, the measured current does not become perfectly constant in the saturation region. Instead, I_D continues to increase slowly as V_{SD} increases. This is consistent with channel-length modulation. Therefore, the saturation-region model used in this lab

includes the factor $(1 + \lambda V_{SD})$.

The extracted value of λ was

$$\lambda = 0.142857 \text{ V}^{-1}$$

and the extracted value of k_p was

$$k_p = 1.314 \times 10^{-3} \text{ A/V}^2$$

The average error between the measured current and the calculated model current was

$$4.38\%$$

The main sources of error include the tolerance of R_D , reading errors from the power supply and multimeter, contact resistance on the breadboard, and the simplified nature of the long-channel MOSFET model.

8) Conclusion

The I_D - V_{SD} characteristics of a PMOS transistor from a CD4007UB chip were measured at

$$V_{SG} = 3 \text{ V}$$

The measured drain resistor was

$$R_D = 989.5 \Omega$$

Using the measured data, the extracted PMOS model parameters were

$$|V_{tp}| = 1.50 \text{ V}$$

$$k_p = 1.314 \times 10^{-3} \text{ A/V}^2$$

$$\lambda = 0.142857 \text{ V}^{-1}$$

The model matched the measured data with an average error of 4.38%. The result shows that the PMOS current-voltage characteristic can be reasonably described by the triode-region equation at low V_{SD} and the saturation-region equation with channel-length modulation at high V_{SD} .

References

- [1] SCUPI 2026S Microelectronics Circuit Lab Manual, Lab #7: PMOS.
- [2] Sedra, A. S., Smith, K. C., *Microelectronic Circuits*, Oxford University Press.
- [3] Texas Instruments, *CD4007UB CMOS Dual Complementary Pair Plus Inverter Datasheet*.